

Universal Radiative Corrections for Precision Determinations of V_{ud}

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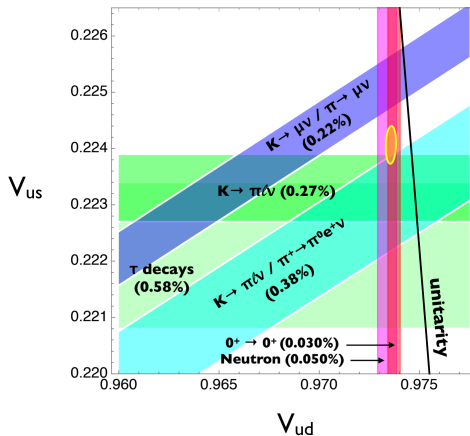
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- ① Semi-leptonic kaon decays
- ② Ratio of V_{us}/V_{ud}
- ③ Radiative corrections and EFT formalism
- ④ NLO EW corrections
- ⑤ NNLO EW corrections

- SM shortcomings: Origin of flavour, CP problem, dark energy, dark matter, matter/anti-matter asymmetry;
- CKM (mixing of quark flavour eigenstates) tested at level 0.01%;
- Cabbibo angle anomaly;
- Measure decay rates: K , $0^+ \rightarrow 0^+$, Π , neutron decay;
- Semileptonic decays: K_{l3} , $0^+ \leftarrow 0^+$, neutron decay;
- Leptonic decays: K_{l2} ;
- Status of V_{us} : Kaon decays;
- Status of V_{ud} : superallowed β decays, neutron decay, pion decays;



$$|V_{ud}|^2 + |V_{us}|^2 - O|V_{ub}|^2 = 1 \quad (1)$$

$$\Gamma(K_{l3}) = \frac{G_F^2 M_K^5}{192\pi^3} C_K^2 S_{EW} |V_{us}|^2 |f_+^{K^0\pi^-}(0)|^2 I_{K\ell}^{(0)} \left(1 + \delta_{SU(2)}^{K\pi} + \delta_{EM}^{K\ell}\right)^2$$

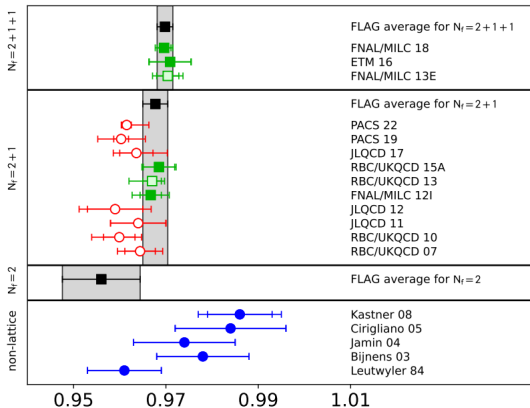
- $S_{EW} = 1.0232(3) \rightarrow$ process independent EW corrections;
- $I_{K\ell}^{(0)}$ phase space in isospin limit;
- $\delta_{SU(2)}^{K\pi}$ strong isospin breaking correction
- $\delta_{EM}^{K\ell}$ long distance EW corrections

$$\langle \pi(p_\pi) | \bar{s} \gamma^\mu u | K(p_K) \rangle = (p_K + p_\pi)^\mu f_+(q^2) + (p_K - p_\pi)^\mu f_-(q^2), \quad (2)$$

Experimental inputs

$$|V_{us}| f_+(0) = 0.21654(41)$$

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 $f_+(0)$ 

$$f_+(0) = 0.9698(17)$$

$$|V_{us}| = 0.22328(58) \quad (K_{\ell 3})$$

$$\frac{\Gamma(K_{\mu 2(\gamma)}^\pm)}{\Gamma(\pi_{\mu 2(\gamma)}^\pm)} = \left| \frac{V_{us}}{V_{ud}} \right|^2 \left(\frac{f_{K^\pm}}{f_{\pi^\pm}} \right)^2 \frac{M_{K^\pm} (1 - m_\mu^2/M_{K^\pm}^2)^2}{M_{\pi^\pm} (1 - m_\mu^2/M_{\pi^\pm}^2)^2} (1 + \delta_{\text{EM}})$$

$$\langle 0 | \bar{q}_1 \gamma^\mu \gamma_5 q_2 | P(p) \rangle = i p^\mu f_P, \quad P = \pi^+, K^+. \quad (3)$$

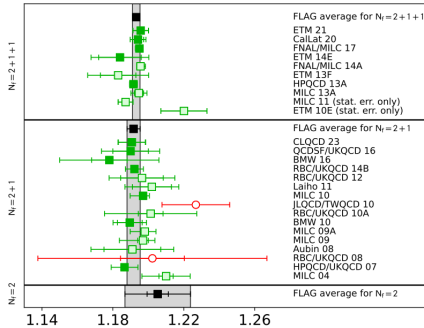
- Average of f_K/f_P in isosymmetric QCD;

Experimental inputs

$$\left| \frac{V_{us}}{V_{ud}} \right| \frac{f_{K^\pm}}{f_{\pi^\pm}} = 0.27599(41)$$

- Isospin correction from the lattice can be improved by higher statistics in contrast with extractions from ChPT;
- Isospin corrections are becoming relevant for CKM elements extraction at the current level of precision of lattice inputs;

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 $f_{K^\pm}/f_{\pi^\pm}$ 

$$|V_{ud}| = 0.97373(31) \quad (\text{superalloved } \beta \text{ decays})$$

$$\frac{f_{K^\pm}}{f_{\pi^\pm}} = 1.1934(19)$$

$$\left| \frac{V_{us}}{V_{ud}} \right| = 0.23126(50) \quad (K_{\ell 2}/\pi_{\ell 2})$$

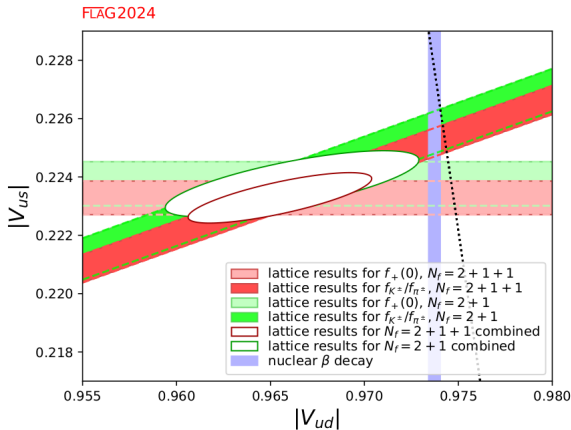


Figure 1: V_{ud} and V_{us} using lattice QCD input and V_{ud} extracted from nuclear β decays.

$$\Gamma_{\pi\beta} = \frac{G_F^2 |V_{ud}|^2 M_\pi^5}{64\pi^3} |g_V^\pi|^2 |f_+^\pi(0)|^2 I_{\pi\ell} (1 + \Delta I_{\pi\ell}) \quad (4)$$

- $f_+^\pi(0)$: pion vector form factor.
- $I_{\pi\ell}$: phase-space integral.
- $\Delta_{\text{RC}}^{\pi\ell}$: radiative corrections.
- g_V^π : effective vector coupling containing the short-distance electroweak corrections.

$$\Delta_{\text{RC}}^{\pi\ell} = 0.03403(11)$$

$$V_{ud}^\pi = 0.97346(283)$$

- The Behrends–Sirlin–Ademollo–Gatto theorem implies

$$f_+^\pi(0) \simeq 1 - 7 \times 10^{-6},$$

making isospin-breaking corrections negligible.

$$\frac{1}{\tau_n} = \frac{G_F^2 |V_{ud}|^2}{2\pi^3} m_e^5 f (1 + 3g_A^2) (1 + \Delta_R), \quad (5)$$

- f : phase-space factor.
- g_A : axial-vector coupling.
- Δ_R : electroweak radiative corrections.

$$\Delta_R^V = 2.436(16)\%,$$
$$|V_{ud}|_n = 0.97410(41).$$

- First NLL calculation of mixed $O(\alpha\alpha_s^2)$ corrections.
- Reduces theoretical uncertainty in Δ_R^V .
- Improves consistency of first-row CKM unitarity tests.

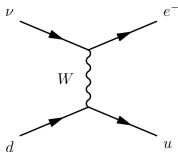


Figure 2: SM tree-level.

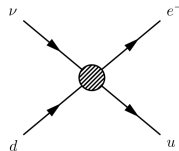


Figure 3: EFT Tree-level.

$$\frac{1}{k^2 - m_W^2} \longrightarrow \frac{-1}{m_W^2} \quad (6)$$

$$H_{eff}(x) = \frac{G_F}{\sqrt{2}} V_{CKM}^i C_i(\mu) Q_i(x) \quad (7)$$

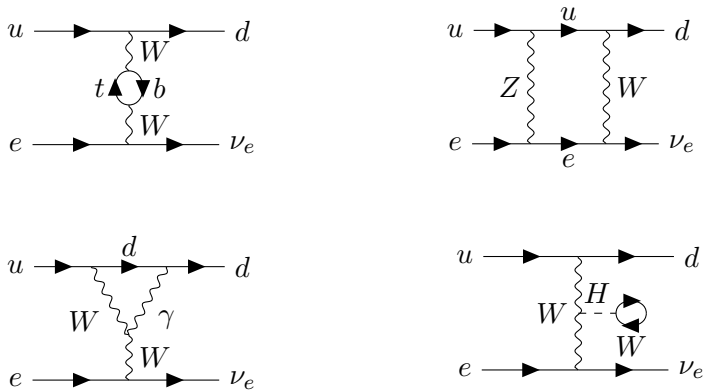


Figure 4: Some 1 loop diagrams in the full theory for the semi-leptonic decay.

Normalization to G_F

$$\mathcal{L}_{G_F} = -\frac{G_F}{\sqrt{2}} Q_\mu - \frac{G_F}{\sqrt{2}} V_{ud} C_q Q_q \quad (8)$$

And the Wilson Coefficients are

$$C_r^{LO} = 1 \quad (9)$$

$$C_r^{NLO} = -5 + \frac{a_\mu^1}{4} + \frac{a_q^1}{12} - 2 \log \left(\frac{\mu^2}{M_Z^2} \right) . \quad (10)$$

- Finite, scheme dependent (a_μ^1 , a_q^1)
- Only box diagrams contribute

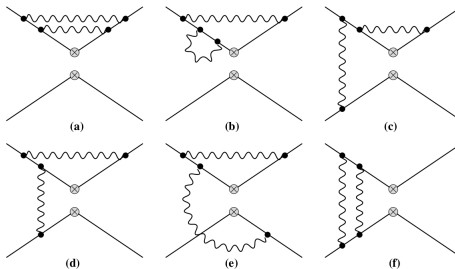


Figure 5: EFT NNLO EW corrections.

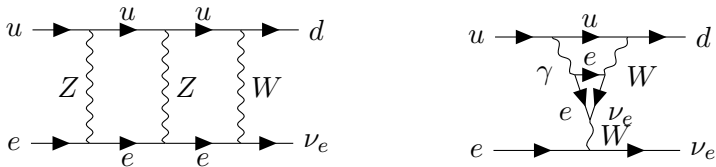


Figure 6: SM NNLO EW corrections.

RGE for WC:

$$\mu \frac{d}{d\mu} C_i^f(\mu) = \sum_j \gamma_{ij}^f C_j^f(\mu), \quad (11)$$

$$C^{(f)}(\mu) = J_f(\mu) U_f(\mu) U_f^{-1}(\mu_0) J_f^{-1}(\mu_0) C^{(f)}(\mu_0) \quad (12)$$

Evolution kernel at LO

$$U_f(\mu) = \left(\frac{\alpha^{(f)}(\mu)}{\alpha(M_Z)} \right)^{-\frac{\gamma^{(0)}}{2\beta^{(0)}}} \quad (13)$$

scheme independence

$$C = U_f^{-1}(\mu_0) J_f^{-1}(\mu_0) C^{(f)}(\mu_0) \quad (14)$$

insert graph with scale dependence

- Combination of lattice inputs and perturbative calculations;
- Higher order calculation + sub-percent lattice calculations;
- Current ongoing work on NNLO EW corrections for V_{ud} extraction from neutron decay.